DAY 08 / 11 · WEB DEV IN 11 DAYS · GUIDED

Real shows from the internet

fetch and APIs

ShowShelf after today: **real shows from TVmaze, with loading and error messages**

TODAY

The plan for the next 90 minutes

Minutes	Topic
10	Recap: five questions about Day 7
10	What is an API?
12	fetch, async and await
13	The shape of the answer, and the adapter
10	Wait, then draw: the loading message
10	When things go wrong
5	Real shows from the start, and the rules
5	The big idea, and questions
15	This afternoon in the Lab

RECAP · DAY 7

Five quick questions

1. querySelector("#grid") finds nothing: what do you get?
2. Three moves put a new element on the page. Which?
3. "input" and "submit": when does each one happen?
4. Why do events change state and call render()?
5. How do you save an array in localStorage, and read it back?

APIS

A website made for programs

`https://api.tvmaze.com`	`/search/shows`	`?q=bad`
the API	what you ask for	the query string

```
const showId = 431;
`https://api.tvmaze.com/shows/${showId}`;   // a template literal builds the address

encodeURIComponent("law & order");          // "law%20%26%20order": safe in an address
```

FETCH, ASYNC AND AWAIT

Ask now, get the answer later

```
async function showOne() {
  const response = await fetch("https://api.tvmaze.com/shows/431");
  const show = await response.json();
  console.log(show.name);   // "Friends"
}

showOne();
```

A **promise** is a ticket: you'll get it later.

await: wait here until it's ready.

await only works inside an **async** function.

THE SHAPE OF THE ANSWER

Every search result is wrapped

results[0].show.name, not results[0].name.

Nothing found? An empty array, [], with status 200.

```
results.map((result) => result.show.id);
```


```
[
  {
    "score": …,
    "show": {
      "id": 169,
      "name": "Breaking Bad",
      "genres": ["Drama", "Crime", "Thriller"],
      "rating": { "average": 9.2 },
      "image": { "medium": "https://…" },
      "summary": "<p><b>Breaking Bad</b> …</p>"
    }
  },
  …
]
```

THE ADAPTER

Their shape → our shape

TVMAZE

```
{
  id: 169,
  name: "Breaking Bad",
  rating: { average: 9.2 },
  premiered: "2008-…",
  network: { name: "AMC" },
  image: { medium: "https://…" }
}
```

toShow()



OURS (DATA.JS SHAPE)

```
{
  id: 169,
  name: "Breaking Bad",
  rating: 9.2,
  premiered: 2008,
  network: "AMC",
  image: "https://…"
}
```

One function translates every show, so createCard and render() don't change.

WAIT, THEN DRAW

Before the await, after the await

```
async function showOne() {
  const title = document.querySelector("h1");
  title.textContent = "Loading…";                                  // runs straight away
  const response = await fetch("https://api.tvmaze.com/shows/169");   // waits here
  const show = await response.json();
  title.textContent = show.name;                                   // runs when it's back
}
```

This afternoon: show the loading message, await the shows, put them in the state, render().

WHEN THINGS GO WRONG

ok, throw, try, catch

No internet: fetch throws an error.

404 (not found) or **500** (server broke): an answer, but response.ok is false.

catch turns any error into a friendly message.

```
async function showOne(id) {
  const title = document.querySelector("h1");
  title.textContent = "Loading…";
  try {
    const response = await fetch(`${API}/shows/${id}`);
    if (!response.ok) {
      throw new Error("TVmaze answered " + response.status);
    }
    const show = await response.json();
    title.textContent = show.name;
  } catch (error) {
    title.textContent = "Sorry: " + error.message;
  }
}

showOne(999999999);   // Sorry: TVmaze answered 404
```

FOUR STATES

Every page that loads data has four states

Loading

“Loading shows…”

Error

A friendly message, in red

Empty

Say what was searched, so a typo shows

Success

The shows

Empty is not an error: TVmaze answers [] with status 200.

THREE MORE THINGS

Real shows from the start, and the rules

Step 9

Page 0

/shows?page=0 lists shows.

No wrapper: each item is a show.

Step 10

Credit

TVmaze's data is free under CC BY-SA.

The page needs a visible link to TVmaze.

Step 11

A new key

Breaking Bad was 2 in the sample data, 169 on TVmaze.

Changed meaning: save under a new key.

THE BIG IDEA

Their data, our shape. Translate it once, as soon as it arrives.

THIS AFTERNOON IN THE LAB

Eleven steps, about three hours

Steps 1–4

Ask TVmaze

1 What is an API?

2 Your first fetch

3 The shape of the answer

4 From their shape to ours

Steps 5–6

On the page

5 Search TVmaze

6 Loading…

Steps 7–8

Problems

7 When things go wrong

8 When nothing is found

Steps 9–11

Finishing

9 Real shows from the start

10 Give credit to TVmaze

11 My List with TVmaze ids

Milestone: the page opens with real TVmaze shows, and a search for friends finds Friends.

STUCK?

In this order

1

Read the message

The “Not yet” box says what to fix.

2

Console

A red line names the problem and the line number.

3

Hint

Two per step. The second is nearly the answer.

4

Answer

Compare it with yours. Find the difference.

5

Ask

Your teacher, or the person next to you.

Most of the time, steps 1 to 3 are enough. That's how developers work too.

END OF THE DAY

Before you go

1. fetch without await: what do you get?
2. Why toShow(result.show), not toShow(result)?
3. The four states of a page that loads data?

FINISHED EARLY?

Extra challenges

Search as you type, politely · Surprise me · How many shows?

TOMORROW

The show page

A page for every show, with its cast and seasons.